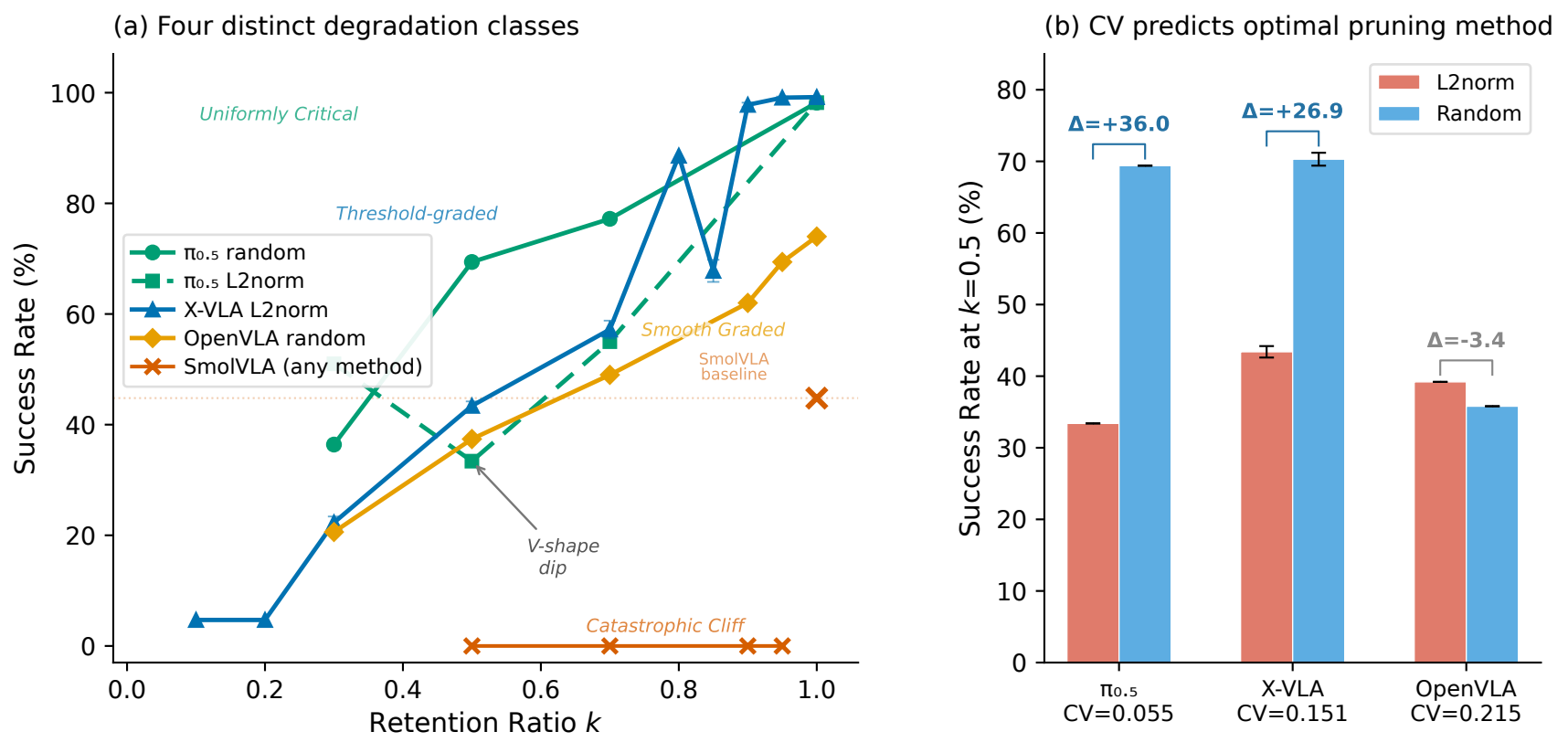

(a) Four distinct degradation classes
Success Rate (%)
Retention Ratio k
Uniformly Critical
Threshold-graded
Smooth Graded
SmolVLA baseline
V-shape dip
Catastrophic Cliff
π0.5 random
π0.5 L2norm
X-VLA L2norm
OpenVLA random
SmolVLA (any method)
(b) CV predicts optimal pruning method
Success Rate at k=0.5 (%)
L2norm
Random
Δ=+36.0
Δ=+26.9
Δ=-3.4
π0.5
CV=0.055
X-VLA
CV=0.151
OpenVLA
CV=0.215